Which AI model is good enough for which task

Measured evidence from Danish upper-secondary physics — a note for the KU AI office

AIPLA project team, Center for Digital Education

2026-08-27

Table of contents

> **Why you are receiving this**
>
> KU is standing up a university-wide AI platform on which staff and students will choose a model appropriate to the task, and has said it intends to look at what that means for the university's energy footprint. Both decisions need the same underlying number: **how good does a model have to be before it is good enough for a given task, and what is the smallest model that clears that bar?**
>
> AIPLA has been measuring exactly that for one subject since spring 2026. This note sets out what we found, what it cost to find out, and where the method transfers to other faculties. The physics numbers will not generalise. **The method, and three of the findings, probably will.**

1. The question

Most public discussion of model choice runs on leaderboards: model X scores 94% on some benchmark, model Y scores 72%, therefore use X. For an institution deciding what to host and what to route where, that framing is close to useless, for two reasons.

First, the benchmarks measure the wrong thing. They are built to separate frontier models at the top of their range — graduate-level science, competition mathematics — and most real university tasks are nowhere near that hard. Summarising a reading, checking a citation format, or helping a first-year student through a mechanics problem does not need a model that can do graduate chemistry.

Second, a leaderboard has no notion of *enough*. The useful question is not "which model is best" but **"what is the cheapest, smallest, most locally hostable model that is still good enough for this particular task?"** We call that the **capability floor** for a task, and it is the number that determines what you buy, what you host, what you route to the cloud, and what your energy bill looks like.

2. What we measured

The task. Solving subquestions from Danish stx *Fysik A* written exams, graded against the official ministry answer keys. This is a real task with a real, defensible ground truth — not a proxy.

The panel. Eleven models on the text task and nine on the figure-reading task, spanning four deployment tiers:

Tier	Where it runs	Data leaves the building?
1	Commercial API (EU regions)	Yes — the prompt does
2	Multi-GPU server or small cluster, self-hosted	No
3	Single GPU or a well-specified workstation, self-hosted	No
4	Laptop, tablet or phone	No

The method. Each model answers into an enforced `{value, unit}` schema with an explicit "I don't know" option, so a text-only model *declines* a question that needs a graph rather than inventing a reading. Grading is a deterministic value-and-unit comparison against the key, plus a language-model judge for the cases where a numeric compare is insufficient. Every model was run **five times**; we report means and standard deviations.

The threshold. We set the bar at **80%** — below that, a tutoring system produces enough confident errors to be a liability rather than a help. The bar is a judgement, and a different task class deserves a different one. Setting it explicitly is the point.

3. What we found

Text: solving exam problems

Means over five runs, 33 text-solvable subquestions from the 2023–24 gold sets:

Model	Tier	Score (mean ± sd)	Clears 80%
Claude (sonnet / opus / haiku)	1 — cloud API	93–96 ± 2	yes
gemini-3.5-flash	1 — cloud API	95 ± 2	yes
deepseek-v4-flash (open)	**2 — multi-GPU, self-hosted**	**91 ± 2**	**yes**
qwen3-32b (open)	**3 — single GPU, self-hosted**	**88 ± 6**	**yes**
gemini-2.5-flash	1 — cloud API	87 ± 3	yes
qwen3-8b (open)	**4 — laptop-class**	**83 ± 3**	**yes, borderline**
qwen3-next-80b-a3b (open)	2	75 ± 6	no
gemini-2.5-flash-lite	1 — cloud API	73 ± 4	no
qwen3-30b-a3b (open)	3	68 ± 9	no
mistral-small-24b (open)	3	67 ± 6	no
gemma-3-27b (open)	3	64 ± 3	no

Figures: reading graphs and diagrams

The questions a text-only model has to decline. Means over five runs, 8 items, vision-capable models only:

Model	Tier	Score (mean ± sd)	Clears 80%
gemini-3.5-flash	1 — cloud API	98 ± 5	yes
qwen3-vl-235b-a22b (open)	**2 — multi-GPU, self-hosted**	**95 ± 6**	**yes**
gemini-2.5-flash	1 — cloud API	93 ± 6	yes
qwen3-vl-32b (open)	**3 — single GPU, self-hosted**	**88 ± 0**	**yes**
glm-4.6v (open)	3	83 ± 6	yes
gpt-5-mini	1 — cloud API	75 ± 0	no
qwen3-vl-8b (open)	4 — laptop-class	63 ± 8	no
gemma-3-27b (open, general-purpose)	3	60 ± 18	no
gemini-2.5-flash-lite	1 — cloud API	40 ± 9	no

The combined picture

A real deployment must handle both kinds of question, so **a tier is only as capable as its weaker modality:**

Tier	Text	Figures	Tier capability	Clears 80%
1 — cloud API	93–96	98	**93**	yes
2 — multi-GPU, self-hosted	91	95	**91**	yes
3 — single GPU, self-hosted	88	88	**88**	yes
4 — laptop / phone	83	63	**63**	no — vision-limited

4. Three findings that may matter beyond physics

Finding 1 — for this task class, self-hosting is already viable

All three server-class tiers clear the bar on both modalities. A **single GPU** running open-weight models scores 88 on both text and figures — comfortably useful, with no data leaving the building and no per-query cost. This is not a projection about 2027; it is a measurement from July 2026.

The corollary matters as much: **the cloud frontier is not required for this task.** Well-posed upper-secondary physics is not frontier-hard. Where a task turns out not to be frontier-hard, paying frontier prices for it is a choice, not a necessity — and the only way to know which tasks those are is to measure them.

The remaining gap is on-device, and it is **figure-reading, not physics**. A laptop-class model already solves the text at the threshold (83); its vision counterpart trails at 63. That gap is bounded by device memory rather than by model quality, so it will close on hardware refresh cycles rather than on model releases.

Finding 2 — a self-hosted tier needs two models, not one

This is the most immediately actionable finding for a model catalogue. The strongest open text models are **architecturally text-only** — they cannot accept an image at all. The strongest open vision-language models *can* do text, but score **12 to 36 points worse** than the text specialist at the same tier, every one of them falling below the threshold. Multimodal training trades away reasoning quality.

So a self-hosted tier serves a text model *and* a vision-language model behind one endpoint, routed by whether the question carries a figure. Provisioning for one model per tier will under-serve one modality or the other, and which one fails will not be obvious from a leaderboard.

Note also that a *general-purpose* open model given an image performs badly and erratically (60 ± 18, with hard failures) where a dedicated vision-language model of the same size is stable at 88 ± 0. "Our model handles images" and "our model handles images well" are different claims.

Finding 3 — published benchmarks will mis-rank your candidates

We tried to shortcut this work by using public benchmark scores, and it did not survive contact with measurement. Three specific failures, each of which would have led to a wrong procurement decision:

- **Some widely-cited figures do not exist.** Scores we found repeated across technical blogs for one major open model turned out to have no basis in its technical report, and appeared in mutually inconsistent versions across sources. We removed them.
- **Vendor self-reported scores implied implausible generational jumps** — up to 42 points over the previous generation's independently verified figure — and had no independent confirmation. On our own task, the models concerned landed far below their published claims.
- **Even honest benchmarks rank the cheap candidates wrongly.** Our scores correlate with the closest public benchmark at roughly r = 0.73 — enough to confirm we are measuring the same underlying capability, not enough to substitute for measurement. The clearest case: the model with the *lowest* public benchmark score of all the open models scored **83** on our task, because the public benchmark is graduate-level and our task is not. A benchmark-based shortlist would have eliminated the cheapest viable self-hosting candidate.

The general form of this: **public benchmarks are calibrated for the top of the range, and most institutional tasks are not at the top of the range.** For any task where the choice actually matters, a small task-specific evaluation beats a large general one.

There is a methodological point attached. Our first version of this table used single runs. Repeating each model five times moved individual scores by up to 15 points and reshuffled the middle of the field, and **open-weight models are markedly noisier than commercial ones**. Any single-run comparison at this resolution — including most of what is published in blog posts — should be treated as unreliable.

5. What this does and does not say about energy

We have measured capability, not energy. But capability per tier is the *precondition* for an energy policy, because it converts a values question into an engineering one.

Once you know that a task class clears its bar on a single self-hosted GPU, then routing that task to a frontier cloud model is a measurable, avoidable cost — and the argument for right-sizing stops being an appeal to restraint and becomes a straightforward efficiency claim. Once you know that a task genuinely needs the frontier, you can pay for it without apology.

What this note supports, then, is a defensible position of the form: *for these task classes we use the smallest tier that clears the bar, and here is the evidence for where that bar sits.* What it does not supply is the energy measurement itself. That would need per-query energy figures for the deployed hardware, which we would be glad to help scope.

6. Limitations — please read before citing

- **Provisional.** Five runs per model, 33 text items and 8 figure items. The rank order and the "tiers 1–3 clear the bar" conclusion are robust to the noise. Individual gaps of less than one standard deviation are not. Read the bands, not the ranks — especially on the figure axis, where a single item is worth 12.5 points.
- **Machine-graded, not yet human-audited.** A physics teacher's calibration sample against a subset of items is the next step and would also pin down how much of the variance is the grader's own.

- **One task class, one subject, one country's curriculum.** Physics exam problems have unusually clean ground truth. Essay feedback, literature synthesis, or code review would each need their own threshold and their own items, and would very likely place the floor at a different tier.
- **Models move.** This is a July 2026 snapshot. The evaluation is versioned and re-runnable precisely because the answer has a shelf life of months.

7. What we can offer

The evaluation is a versioned dataset plus a runner, built to be re-run as new models release, and it is not physics-specific in its machinery — only in its items.

- **The current results and the runner**, for KU's own model-catalogue work.
- **Help defining a capability floor for another faculty's task classes.** The expensive part is not the software; it is deciding what "good enough" means for a given task and assembling thirty or so items with defensible ground truth. That is roughly a week of a domain expert's attention per task class, and it is the sort of concrete, bounded piece of work the AI-labs appear designed to support.
- **A walkthrough** with whoever is deciding the model catalogue.

AIPLA's technical lead is engaged with the project to at least April 2027 and based at the Center for Digital Education. The underlying evaluation is a research instrument intended to outlive the current work.

> The single sentence
>
> If KU is going to serve models centrally and route tasks to the smallest adequate one, then somebody has to decide what "adequate" means for each kind of task — and that decision is measurable, cheap to measure, and wrong if taken from a leaderboard.